Mandatory Assignment 1

Your names

1 Quarto: a framework for reproducible research

Quarto weaves text and executable code into one document. To learn more, see https://quarto.org and the course's Quarto self-study page. When you click **Preview** in Positron or run `quarto render` in the Terminal, Quarto runs the code and creates a PDF with all results. Use this template for your mandatory assignments. It shows the elements every report needs: sections, a figure and a table with captions and cross-references, math, a footnote, citations, and numbers in the text that come from the code (inline code). By default, the PDF hides the code; `echo: true` in the header or in a chunk shows it.

2 Exercise

1. Download daily adjusted prices for three stocks, AAPL, MSFT, and BA, from January 1, 2000, until December 31, 2025, from Yahoo Finance. Show the code you use to download the data in your report. Compute the daily returns of each stock.

```
# Daily prices from Yahoo Finance, adjusted for splits and dividends
prices = tf.download_data(
    domain="Stock Prices",
    symbols=symbols,
    start_date=start_date,
    end_date=end_date,
)
```

2. Visualize the distribution of the daily returns.

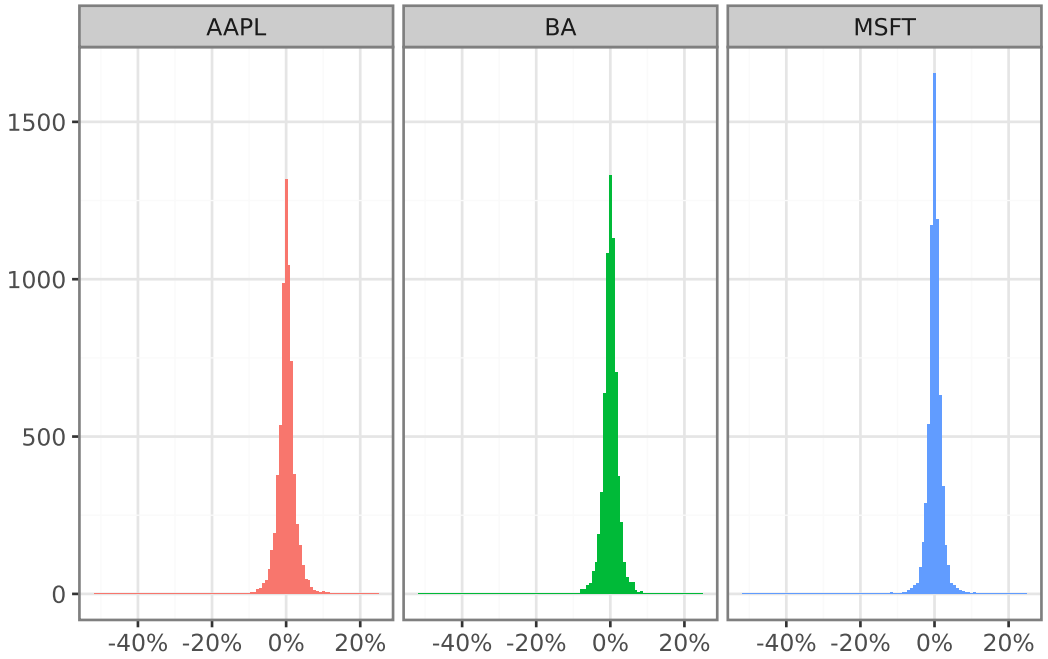


Figure 1: Distribution of daily returns of AAPL, BA, and MSFT, January 2000 to December 2025.

Figure 1 shows the distribution of daily returns for the three stocks.

3. Compute the sample mean μ and the variance-covariance matrix Σ of the returns. Which of the individual assets delivered the highest Sharpe ratio (assume a risk-free rate of zero) during the sample period?[1] The estimated moments may later be used to construct optimal portfolios (Markowitz, 1952; Merton, 1972).

Answer: We compute the annualized Sharpe ratio of asset i as

$$\sqrt{252}\,\frac{\hat{\mu}_i}{\hat{\sigma}_i},$$

where $\hat{\mu}_i$ is the sample mean and $\hat{\sigma}_i$ the sample standard deviation of asset i's daily returns, and 252 is the number of trading days per year. AAPL has the highest annualized Sharpe ratio, 0.779. Table 1 shows the Sharpe ratios of all three stocks.[2]

Table 1: Annualized Sharpe ratios of daily returns, January 2000 to December 2025.

Ticker	Sharpe ratio
AAPL	0.779
BA	0.409
MSFT	0.485

[1] Whenever you report performance measures, compute annualized values and state clearly how you calculated them.

[2] Format numbers for the text with an f-string, as here. Otherwise Quarto prints Python's raw representation, for instance `np.float64(0.776)`.

References

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Merton, R. C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4):1851–1872.